MILITARY SPECIFICATION

SOLENOID, FIRING, CALIBER .50 GUN, 28 VOLT DC, TYPE G-22A

1. SCOPE

1.1 This specification establishes the requirements for one type of caliber .50 gun electrical firing solenoid, designated as type G-22A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-637	- Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Box, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-C-45662	- Calibration System Requirements.
MS20073	- Bolt, Machine, Aircraft, Drilled Head, Fine Thread.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized and Containerized Unit Loads 40"x48" Pallets, Skids, Runners, or Pallet-Type Base.

DRAWINGS

US AIR FORCE

49C4267	- Solenoid Assembly - Gun, Type G-22A.
45A16979	- Bar Assembly - Solenoid Trigger.
45A16987	- Screw - Solenoid Mounting.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.)

3. REQUIREMENTS

3.1 First article. First article solenoids shall be furnished for the inspections specified in 4.5. When specified in the contract or purchase order (see 6.2), the following data shall be supplied with the first article sample:

(a) Three copies of outline dimensional sketches.
(b) Three copies of practical wiring diagram.
(c) Prior to submission of the first article sample equipment, three copies of a report by the contractor of his tests on the equipment. Included in this shall be an analysis of all failures which occurred.

3.2 Drawings. The solenoid requirements shall be as specified herein and in accordance with US Air Force Drawings 49C4267, 45A16979, and 45A16987.

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the solenoids to meet the performance requirements of this specification. Materials used shall be self-extinguishing, and shall not support combustion, given off noxious gases in harmful quantities, give off gases in quantities sufficient to cause explosion of sealed enclosures, cause contamination to any part of the solenoid, or form current carrying tracks when subjected to any of the tests specified herein. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion.

3.5 Design and construction. Solenoids shall be of the design, construction, and physical dimensions specified. Solenoids shall be so designed to insure proper operation when mounted in any position. Solder shall not be used primarily for obtaining mechanical strength. All flux shall be removed after soldering.

3.5.1 Threaded parts. All threaded parts shall be in accordance with Handbook H28. Where practical, all threads shall be in conformity with the coarse-thread series. The fine-thread series may be used only for applications that show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch between 16 and 36, which is used in the fine-thread series.

3.5.2 Trigger bar. One trigger bar assembly, meeting the requirements of US Air Force Drawing 45A16979, shall be furnished with each solenoid.

3.5.3 Mounting screw and bolt. One mounting screw, meeting the requirements of US Air Force Drawing 45A16987, and one mounting bolt, meeting the requirements of MS20073, shall be furnished with each solenoid.

3.6 Electrical requirements.

3.6.1 Current consumption. When solenoids are tested as specified in 4.7.2, the current consumption shall not exceed 6.2 amperes at 28 volts dc.

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3.6.2 Operation. When solenoids are tested as specified in 4.7.3, the solenoid shall develop a minimum static effort of 120 pounds at an air gap of .022 inch.

3.6.3 Time of operation. When solenoids are tested as specified in 4.7.4, the time required for the solenoid plunger to travel from its fully open position to its fully closed position shall not exceed 0.04 second, and also that the time required for the solenoid plunger to travel from its fully closed position (air gap closed) to its fully open position, after breaking the circuit, shall not exceed 0.04 second.

3.6.4 Temperature rise. When solenoids are tested as specified in 4.7.5, there shall be no damage to the solenoid as a result of this test.

3.6.5 Immersion. When solenoids are tested as specified in 4.7.6, there shall be no evidence of mechanical or electrical damage.

3.6.6 Kinetic energy. When tested as specified in 4.7.7, the solenoid shall develop a minimum of one tenth (.10) foot-pound kinetic energy.

3.6.7 Gun fire. Solenoids shall be forwarded to the procuring activity for the gun fire test.

3.7 Marking. Solenoids shall be marked in accordance with MIL-STD-130, with the following information:

SOLENOID, FIRING, CALIBER .50 GUN
TYPE G-22A
28 VOLTS DC
MFGR'S PART NUMBER _____*
MFGR'S NAME OR TRADEMARK _____*
US PROPERTY

* Applicable data to be entered by the contractor.

3.8 Workmanship. The solenoids shall be fabricated in such a manner as to be uniform in quality, and shall be free from cracked or displaced parts, sharp edges, burrs, and other defects that will affect life, serviceability, and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.4).
(b) First article inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

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4.3 Inspection conditions. The following conditions shall apply during testing of current consumption, operation, time of operation, temperature rise, and immersion (see 4.7.2 to 4.7.6 inclusive).

Operating voltage	28 volts dc.
Temperature	Room ambient 25° $^{+15°}_{-5°}$ C.
Altitude	Normal ground.
Humidity	Room ambient up to 90 percent relative humidity.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the solenoids are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.5 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.5.1 Sample size. Unless otherwise specified (see 6.2), one solenoid shall be subjected to first article inspection.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown.

4.5.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

TABLE I. Inspection of solenoids.

Examination or test	Requirement paragraph	Method paragraph
Group I		
Visual and mechanical examination	3.1, 3.2, 3.3, 3.4, 3.5. 3.7, and 3.8	4.7.1
Current consumption	3.6.1	4.7.2
Operation	3.6.2	4.7.3
Group II		
Time of operation	3.6.3	4.7.4
Temperature rise	3.6.4	4.7.5
Immersion	3.6.5	4.7.6
Kinetic energy	3.6.6	4.7.7
Group III		
Gun fire 1/	3.6.7	4.7.8

1/ The gun fire test will be conducted by the procuring activity, due to the lack of firing ranges at most manufacturing plants.

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4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups I, II, and III of table I.

4.6.2 Inspection lot. An inspection lot shall consist of all the solenoids utilizing the basic manufacturing process requirements, produced under essentially the same conditions, and offered for inspection at one time.

4.6.3 Sampling plan. Group I tests shall be performed on 100 percent of the solenoids furnished under this specification; group II tests shall be performed on four solenoids from each inspection lot; for group III tests, one solenoid out of the first 100 shall be subjected to a 10,000-round gun fire test. One solenoid out of each successive lot of 3,000 shall be subjected to a 2,500-round test. No failures shall be allowed.

4.6.4 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit four additional samples for inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.5 Noncompliance. If a sample fails to pass quality conformance inspection, the supplier shall take corrective action on the materials or processes, or both as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, quality conformance inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Final acceptance shall be withheld until the quality conformance inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failures and corrective action taken shall be furnished to the cognizant inspection activity.

4.6.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination (external). Solenoids shall be examined to verify that the materials, external design and construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.2, 3.3, 3.4, 3.5, 3.7, and 3.8).

4.7.2 Current consumption (see 3.6.1). Solenoids shall be tested to determine that the current consumption does not exceed the value specified in 3.6.1 when 28 volts dc is applied.

4.7.3 Operation (see 3.6.2). Solenoids shall be tested at a starting and ambient temperature of 20° C (68° F) and with a constant EMF of 28 volts dc applied. The solenoid shall develop the static effort specified in 3.6.2.

4.7.4 Time of operation (see 3.6.3). Solenoids shall be tested at a starting and ambient temperature of 20° C (68° F) and with a constant EMF of 28 volts dc. The solenoids shall meet the requirements specified in 3.6.3.

4.7.5 Temperature rise (see 3.6.4). Solenoids shall be subjected to a constant EMF of 28 volts dc for 5 minutes in still air at a starting and ambient temperature of 20° C (68° F). Then allow solenoids to cool to the starting temperature. Following this test there shall be no mechanical damage and the solenoid shall satisfactorily pass the following tests:

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Current consumption - - - - - As specified in 4.7.2.
Operation - - - - - - - - - - - As specified in 4.7.3.
Time of operation - - - - - - As specified in 4.7.4.

4.7.6 Immersion (see 3.6.5). Solenoids shall be completely immersed in fresh water for a period of 24 hours. Following immersion, solenoids shall be dried and shall satisfactorily pass the following tests:

Operation - - - - - - - - - As specified in 4.7.3.
Time of operation - - - - As specified in 4.7.4.

4.7.7 Kinetic energy (see 3.6.6). Solenoids shall be tested in the following manner and shall conform to the requirements of 3.6.6:

(a) Clamp solenoid firmly in inverted position and supply a 28 volt dc source with a test switch to the solenoid AN connector.
(b) Attach fixture (see figure 3) to the solenoid as shown in figure 1.
(c) Place ball (see figure 2) carefully on top of the plunger with the flat side resting on the plunger so that it will not slide off.
(d) Close test switch and observe that the ball strikes the test plate. The plate should ring audibly when struck by the ball. If the ball does not strike the plate, the solenoid has not developed enough energy to be accepted.

4.7.8 Gun fire (see 3.6.7). Solenoids shall be tested in accordance with the requirements of 3.6.7, on a free firing caliber .50 M3 machine gun. This test shall in no way indicate damage to the unit.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Solenoids shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Dryi Solenoids shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Solenoids shall be individually packaged in accordance with MIL-P-116, submethod IIc insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Each unit package shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Cleaned and dried solenoids shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged solenoids shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using non-metallic or tape banding only.

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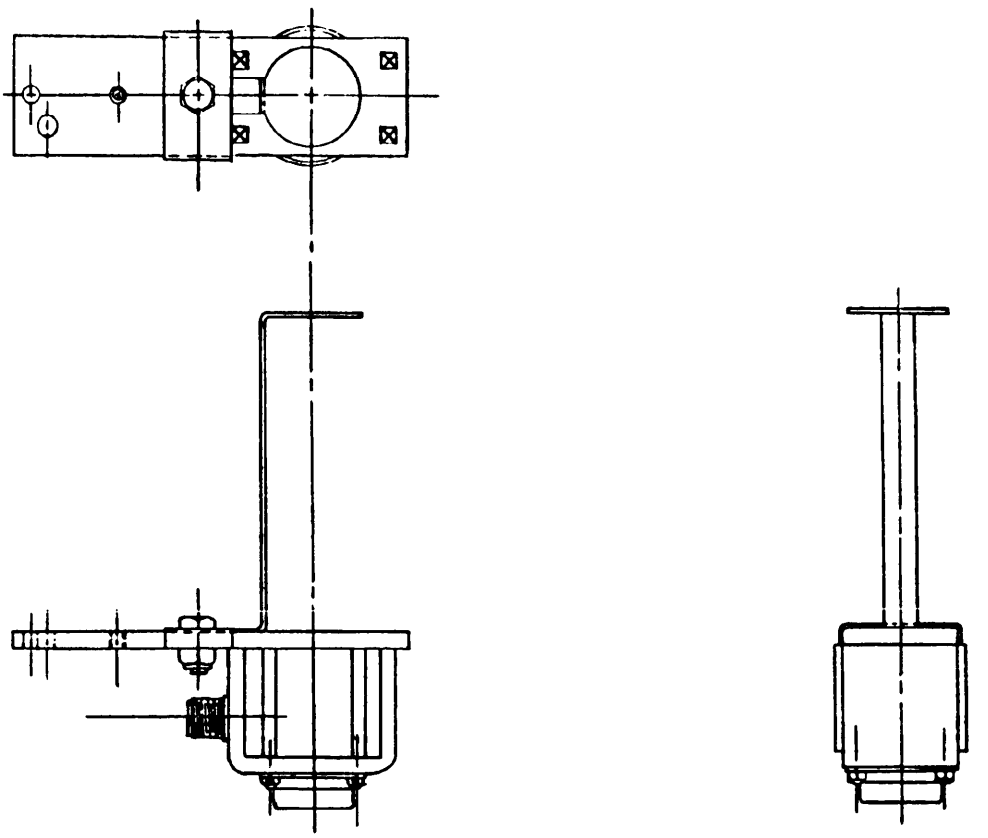
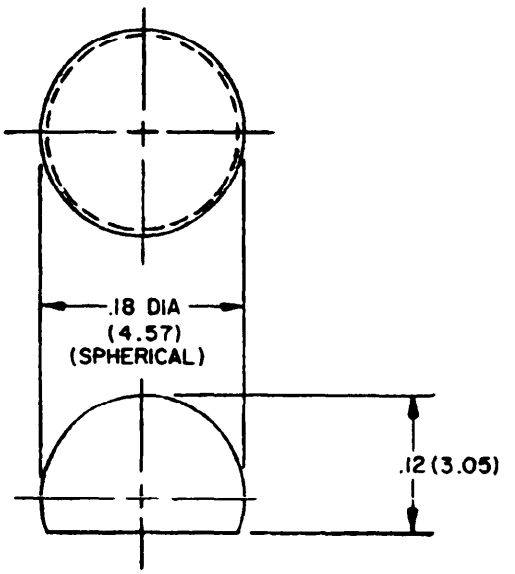


FIGURE 1. Method for attaching fixture to solenoid.

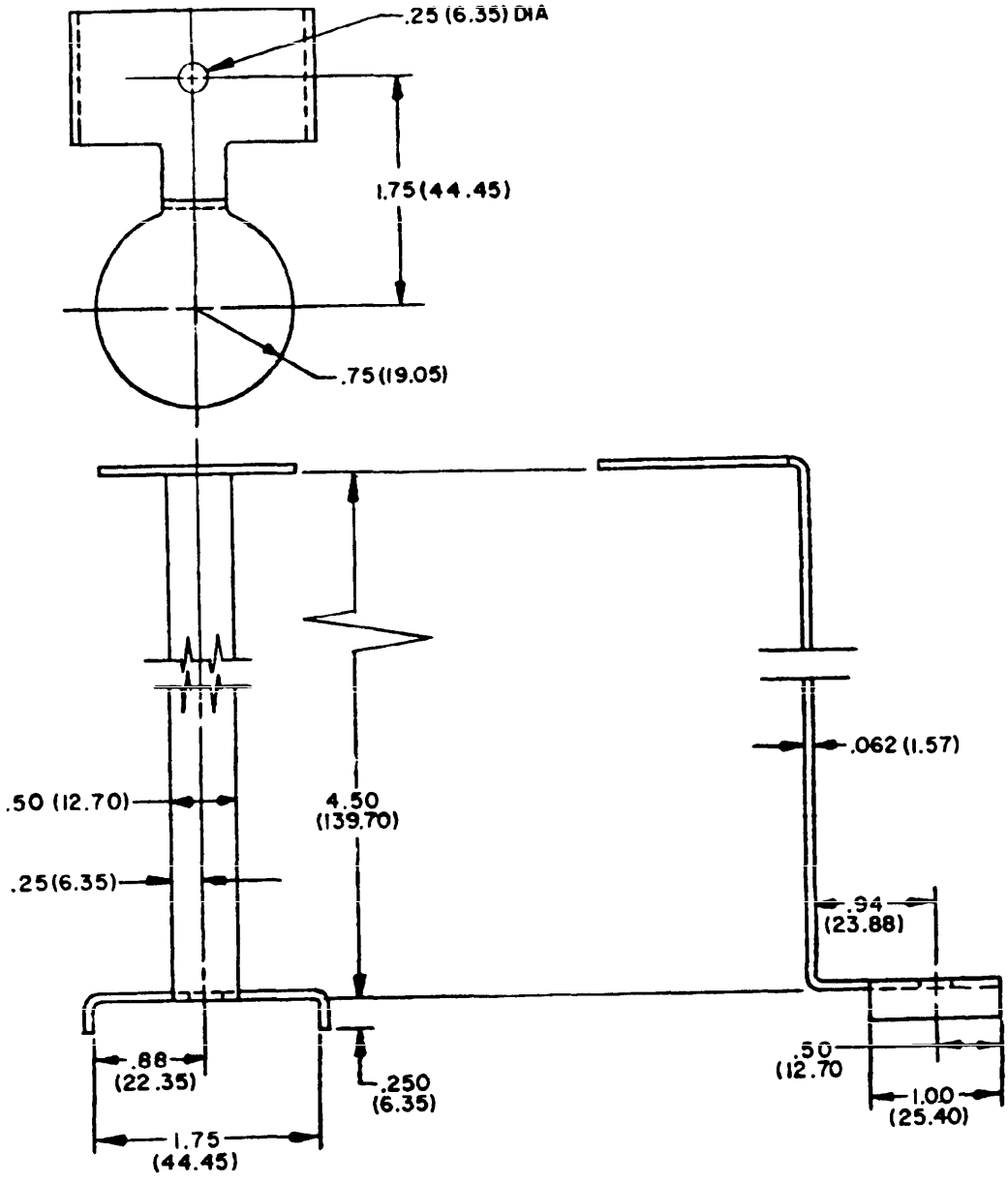
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Test ball is to be fabricated of brass spec. QQ-B-637, class A, 1/2 hard.

FIGURE 2. Test ball.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Part is to be fabricated from sheet steel spec. QQ-S-698 cond. 2 dull finish.
5. All bend and relief radii .06(1.52 mm).

FIGURE 3. Fixture.

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5.2.2 Level B. The packaged solenoids shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged solenoids shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Solenoids, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Solenoids, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Solenoids, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, supplementary and exterior container and unitized load shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Solenoid, Firing, Caliber .50 Gun, 28 Volts DC, Type G-22A
Specification MIL-S-7014C(USAF)
Mfr's Part No. ____*____

* Applicable data shall be entered by the supplier.

5.4 General. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

6. NOTES

6.1 Intended use. Type G-22A solenoid as covered by this specification is intended for use in aircraft for the electrical control of free firing caliber .50 machine guns.

6.2 Ordering data. Requisitions, purchase orders, and contracts should state the name, type, and model designation of the equipment; the title, number, and date of this specification; and list the components by name and type designations as appearing on the current parts control list; whether overseas or domestic packing is desired; and the number of first article samples to be inspected and data required.

PATENT NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data, is not be be regarded by implication or otherwise as in any manner licensing the holder or other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention thay may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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